

Ayush Amin

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EDUCATION

Vellore Institute Of Technology

Bachelor of Technology in Computer Science - **90.1%**

Vellore, India

Sep. 2022 – July 2026

TECHNICAL SKILLS

Technical Skills: Golang, Python, JavaScript, TypeScript, Backend Development, Distributed Systems, REST APIs, System Design, Concurrency, Caching (Redis), Search Systems (Elasticsearch), Docker, MongoDB, SQL

AI & LLM Systems: RAG Pipelines, Agentic AI Workflows, Prompt Engineering, LLM Orchestration, Vector Databases (ChromaDB/FAISS), Embeddings, Context Management, Evaluation & Retrieval Optimization

Frameworks: Go (net/http, goroutines, channels), Gin, Fiber, Echo, FastAPI, Node.js, Next.js, LangChain, LangGraph

Soft Skills: Problem-solving, Time Management, Communication, Teamwork, Decision Making

EXPERIENCE

TIFIN

Software Engineer FTE

May 2026 – Present

Bangalore, India

- Built and fine-tuned the organization's core financial LLM powering **100%** of B2B and B2C AI workflows across production services
- Developed an AI-powered Slack agent integrated with GitHub, Linear, GCS, and Codemagic MCPs, automating PR reviews, deployments, build generation, and ticket workflows
- Engineered and deployed GPU health monitoring endpoints for **L40** and **H100** clusters with automated failure detection and Slack-based alerting, reducing downtime response latency by **65%**
- Designed an automated LLM evaluation framework using Google Cloud Scheduler to classify failures into LLM, data, or infrastructure issues with query-level diagnostics and ownership tagging

MyFi by TIFIN

Automation & Developer Intern

Dec 2025 – April 2026

Bangalore, India

- Built a RAG-based insurance policy analysis pipeline using GPT-4o-mini vision, enabling real-time Q&A over PDFs and reducing manual review effort by **60%**
- Engineered an asynchronous document pipeline enabling real-time querying of partially processed documents, improving throughput by **50%**
- Developed backend systems with Redis caching and GitHub webhook processing, reducing MongoDB read load by **60%** and improving deployment tracking

Accenture

Advanced App Engineering Analyst Intern

May 2025 – July 2025

Hyderabad, India

- Connected backend services with testing frameworks to ensure **99%** system reliability across environments
- Executed end-to-end testing and resolved critical bugs, improving release quality and system stability
- Optimized a Next.js 15 feature with Partial Pre-rendering (PPR), reducing page load time by **35%**
- Deployed services on Google Cloud Platform (GCP), improving scalability and deployment efficiency
- Authored technical and functional documentation for final delivery to engineering and QA teams

PROJECTS

AI-Research | Multi-Agent LLM, RAG, Go, FastAPI, Ollama, ChromaDB

GitHub Link

- Architected a local-first multi-agent research system orchestrating **6 specialized LLM agents** (Planner, Search, RAG, Summarizer, Validator, Formatter) to produce evidence-backed reports from a single user query
- Engineered an async streaming pipeline between a **Go API server** and a **Python FastAPI orchestrator**, relaying NDJSON agent events over Redis-backed SSE for real-time visibility into every reasoning step
- Built a hybrid retrieval layer combining **ChromaDB vector search** with live web evidence, running Search and RAG agents in parallel via **asyncio.gather** to cut end-to-end research latency by **45%**
- Hardened LLM reliability with structured Pydantic schemas, planner fallback paths, and a validator agent that flags low-evidence runs, achieving **100% on-prem inference** via Ollama (Qwen) with zero external API dependency

CloudReach.AI | Python, RAG, LangChain, Llama3

GitHub | Live

- Built a RAG-based system to generate personalized cold emails, reducing manual effort by **70%**
- Designed a retrieval pipeline with chunking, embeddings, and semantic search, improving relevance by **35%**
- Reduced generation time from **5 minutes to under 20 seconds** via optimized prompt and context handling
- Improved personalization using HuggingFace models, increasing response rates by **18%**